



sail away

pl > bachotex 2026

de > context 2026

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prompting donald knuth

memory lane

Stanford has videos online where Knuth presents and discusses T_EX development. Watching them brings me back to school:

- artificial intelligence was discussed: expert systems (prolog)
- there were experiments with computer assisted teaching: experimental authoring systems
- there was little clue how to apply computers in education (the apple 2 network, atari, m amiga were used for gaming)
- we ended up with: cai language (vm based), controlling video tapes (random access), simulations
- computers were not yet everywhere, there was little interest, and it was time consuming

some observations

- 2 >> 1. case insensitive control sequences
- 2 >> 2. limited identifier length might have inspired hashing
- 2 >> 3. did the language used influence the implementation
- 2 >> 4. it started out without hash but with strings and lookups
- 2 >> 5. the 7 bit limitations were later removed
- 2 >> 6. subtle changes took place, like spaced after single char `\cs`

The numbers refer to fragments in `musings-basic.pdf` (available in the ConTeXt distribution) that also has more details.

more observations

- 2 >> 7. it started with 32 fonts and \: to change one
- 2 >> 8. they were tagged with single ascii characters
- 2 >> 9. grouping with { } was used to limits the scope
- 2 >> 10. the nesting level was actually more than we need today
- 2 >> 11. the par builder evolved over time
- 2 >> 12. spaces after macro bodies were ignored

even more observations

2 >> 13. most units were there, but things changed a bit

2 >> 14. hyphenation was hard-coded

2 >> 15. tracing and logging was very much related to terminals

2 >> 16. math spacing was there but much was frozen

the basic format

- 3 >> 1. special symbols (like arrows for assignments) are used
- 3 >> 2. we don't have a variables like `\hsize` yet
- 3 >> 3. there is a concept of ragged right built in
- 3 >> 4. commands like `\sin` are curiously coded
- 3 >> 5. the `\quad` primitive is hard coded
- 3 >> 6. there are various hard codes kern commands (again symbols)

more basics

- 3 >> 7. there is already a fight going on with spurious spaces
- 3 >> 8. the various fonts show up (not yet c-named)
- 3 >> 9. in addition to single char font changes we have `\rm` etc.
- 3 >> 10. we already have parameters that control spacing
- 3 >> 11. but `\textfont` etc. definitions are more obscure
- 3 >> 12. we have some counters but they behave differently

start sailing

- 4 >> 1. the first version was written in sail, a kind of pascal
- 4 >> 2. it definitely didn't look like the later web'd source
- 4 >> 3. the cmd plus chr 'instruction' model is visible
- 4 >> 4. the meaning of characters (catcodes) is hard coded
- 4 >> 5. concepts have different 'tags' (block for group)
- 4 >> 6. extensions have always been there (!)

sail further

4 >> 7. there is talk about 'reals' (no registers yet)

4 >> 8. advancing a counter is limited by step one

4 >> 9. conditionals are kind of odd (and \ifeven)

4 >> 10. two classes of inserts are hard coded (Don dislikes notes)

4 >> 11. box sized can use 'size', no \hsize yet

4 >> 12. we have marks but no first one yet

sail away

4 >> 13. instead of special penalties we have \eject

4 >> 14. equations are always numbered right

4 >> 15. math concepts like fences and fractions will later evolve

4 >> 16. hanging is there but different, and no shapes yet

4 >> 17. the eqtb[...] entries are there but way less

4 >> 18. glues are already nodes and much is hardcoded

the intermezzo

- 5 >> 1. there were some changes between the major releases
- 5 >> 2. registers were introduced and \advance evolved
- 5 >> 3. it became clear that conditionals had to be redone
- 5 >> 4. units matured and box dimensions got primitives
- 5 >> 5. catcodes, char codes and math codes showed up
- 5 >> 6. glue got order and fillers, no more extreme values

post sail tex

1. the early '82 version was major refactoring
2. there came a better organization of the 'virtual machine'
3. hyphenation, par and page building, math improved
4. inserts, marks and adjusts grew up
5. we got more registers and more internal parameters
6. metafont became quite different

post sail tex

7. documenting in web for sure made us end up here
8. testing and feedback lead to subtle additions
9. between '82 and '89 we see some cleanup
10. 7 bit became 8 bit

timeline

1. the errorlog.tex file gives a good overview
2. (the smalltalk project has a similar paper trail)
3. all this helps 'a history of computer science'
4. tex software evolved alongside the hardware
5. the videos and books show the teamwork
6. it took a Donald Knuth to pull this off